

One visible parent: one answer?

Learning intention

Distinguish sexual and asexual plant reproduction using process evidence rather than the number of visible parent plants.

A new plant grows near one parent.

Can you identify the reproduction type from that alone?

Recover the fusion event

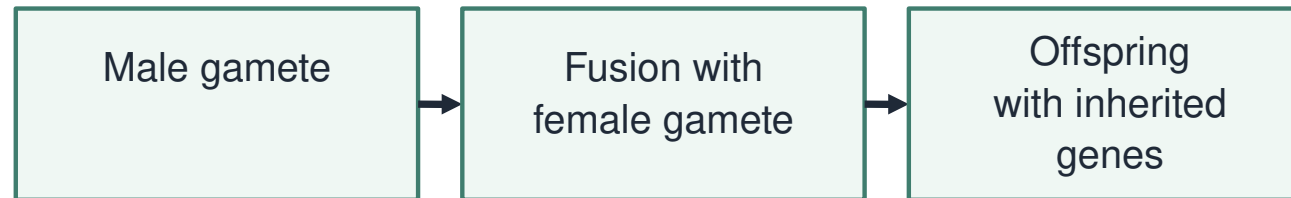
R1. What is fertilisation?

R2. What develops into a seed after fertilisation?

R3. Is growth of one plant the same as producing offspring?

I look for gamete fusion

Sexual reproduction: the defining event



Gametes may come from different plants
or from one self-fertile plant.

Fusion identifies sexual reproduction. Gametes can come from one self-fertile plant. Two separate plants are not the defining rule.

I trace the runner route

A runner can produce a daughter plant



Vegetative growth from the parent; no gamete fusion here.

The new plant grows from a parent stem. No gametes fuse in this runner route. This is asexual reproduction.

I separate genetic similarity from appearance

Clones need not look identical



Illustrative model: genes and growth conditions both affect traits.

A cutting can produce a clone of its source. Conditions can affect size and appearance. “Clone” does not mean every visible feature is fixed.

Nursery case conference

W1. Sort A–D using gamete fusion. Give the evidence phrase that decides each case. Do not count parent plants alone.

Case	Key evidence
A: crossed flowers	Gametes fuse
B: self-fertile flower	Gametes fuse
C: runner	No gamete fusion
D: cutting	No gamete fusion

Choose a plan for the nursery

W2. The nursery wants inherited mint features retained.

Recommend the stated cutting route or a new cross.

Give one reason and one condition to control.

Challenge the shortcut rules

W3. Repair “one parent always means asexual”.

Then repair “all clones always look identical”.

A flower self-fertilises

Its male and female gametes fuse. Which route is this?

- A** Sexual reproduction.
- B** Asexual because only one parent plant is visible.
- C** No reproduction, only taller growth.

Your independent investigation

Choose one route.

Use the stated process to classify each case.

Explain a nursery decision and its limits.

Use the evidence

Underline “gametes fuse” or “no fusion”.

Name runner or cutting when relevant.

Separate genes from growth conditions.

Check before sharing

Does my explanation answer the question?

Have I named the evidence or process?

Have I kept the conclusion within its limits?

Compare and improve

Share one classification and reason.

Check a broad rule against the case cards.

Improve an absolute clone claim.

A question worth investigating

Suggest one next question about this lesson's evidence.

Choose a question whose answer would improve our model.

Show the science you can explain

- E1. What distinguishes sexual reproduction?
- E2. Give one asexual plant method.
- E3. Can self-fertilisation be sexual? Explain.